Entanglement cost of every two-qubit state

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Abstract

We establish strong superadditivity of entanglement of formation for arbitrary finite-dimensional mixed states with a distinguished two-qubit subsystem. Consequently, entanglement of formation is additive for every two-qubit state paired with any finite-dimensional bipartite state, and the asymptotic entanglement cost equals the Wootters expression. Our proof combines a balanced rectangular block-matrix entropy inequality derived from the Schatten–Hanner inequality with a transverse-correlation bound for concurrence at fixed marginal eigenaxes.

1. Definitions and result

All Hilbert spaces are finite dimensional. Tensor products and partial traces are those of ordinary quantum mechanics; no superselection restriction is imposed. All entropies are in bits, except in the explicitly marked differentiation calculation. For a bipartite state,

$$E_F(\rho_{A:B}) = \inf_{\rho=\sum_j q_j|\psi_j\rangle\langle\psi_j|} \sum_j q_j S(\mathrm{Tr}_B\,|\psi_j\rangle\langle\psi_j|).$$

Write $h_2(t) = -t\log_2 t - (1-t)\log_2(1-t)$, with $0\log 0 = 0$, and

$$W(c) = h_2\left(\frac{1+\sqrt{1-c^2}}{2}\right), \qquad 0 \le c \le 1.$$

For a two-qubit state let $J = \sigma_y \otimes \sigma_y$, $\tilde{\rho} = J\rho^* J$, and let $\lambda_1 \ge \cdots \ge \lambda_4$ be the square roots of the eigenvalues of $\rho\tilde{\rho}$. Wootters' theorem [1] states

$$C(\rho) = \max(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4), \qquad E_F(\rho) = W(C(\rho)).$$

We use two further consequences established in the same work: concurrence is the convex roof of pure-state concurrence and hence is convex, and W is increasing. Product ensembles imply the usual subadditivity of E_F.

Theorem. Let A, B be qubits and let A', B' have arbitrary finite dimensions. Every state $\Omega_{AA'BB'}$ satisfies

$$\boxed{E_F(\Omega_{AA':BB'}) \ge E_F(\Omega_{A:B}) + E_F(\Omega_{A':B'}).} \tag{1}$$

Consequently, for any two-qubit state ρ and any finite-dimensional bipartite state σ,

$$E_F(\rho \otimes \sigma) = E_F(\rho) + E_F(\sigma). \tag{2}$$

In particular, for all two-qubit states,

$$\boxed{E_F(\rho^{\otimes n}) = nW(C(\rho)) \quad \text{for every } n \geq 1.} \tag{3}$$

The operational entanglement cost under unrestricted LOCC, free classical communication, vanishing trace-distance error and no extra nonlocal resource is therefore

$$\boxed{E_C(\rho) = W(C(\rho)).} \tag{4}$$

The proof uses a balanced block entropy inequality and a fixed-axis concurrence bound. Both are proved below. The substantial imported results are the Schatten Hanner inequality [2], the Wootters formula [1], and the operational regularization theorem [3].

2. Balanced block entropy

Lemma 1. Let M be a complex $2r$-by-$2s$ matrix of Frobenius norm one, partitioned into r-by-s blocks M_{ij}. Put $p_{ij} = \|M_{ij}\|_2^2$, using the Schatten 2 norm. Suppose

$$p_{00} = p_{11} = x/2, \qquad p_{01} = p_{10} = y/2, \qquad x + y = 1.$$

Then

$$S(MM^\dagger) \geq W(|x - y|) + \sum_{i,j} p_{ij} S(M_{ij} M_{ij}^\dagger / p_{ij}). \tag{5}$$

A zero-weight term is defined to be zero.

Proof

Heinävaara's Corollary 1.1 [2] gives, for arbitrary complex matrices X, Y and $q \geq 2$,

$$\|X + Y\|_q^q + \|X - Y\|_q^q \leq (\|X\|_q + \|Y\|_q)^q + |\|X\|_q - \|Y\|_q|^q. \tag{6}$$

The complex-matrix scope is explicit in Corollary 1.5. Rectangular matrices are included by zero-padding, which preserves singular values. No positive-semidefinite or Hermitian assumption on X, Y is needed.

Set

$$X = \begin{pmatrix} M_{00} & 0 \\ 0 & M_{11} \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & M_{01} \\ M_{10} & 0 \end{pmatrix}.$$

The matrix $X - Y$ is obtained from $M = X + Y$ by changing the signs of the lower block row and right block column. It has the same singular values as M. Thus (6) becomes

$$2\|M\|_q^q \leq (r(q) + s(q))^q + |r(q) - s(q)|^q, \tag{7}$$

where $r(q) = \|X\|_q$, $s(q) = \|Y\|_q$. Both sides equal 2 at $q = 2$.

Use natural logarithms for the next calculation, writing S_e and $W_e = (\ln 2)W$. For any fixed nonzero matrix Z, with $z = \|Z\|_2^2$, differentiation of its singular-value sum gives

$$\frac{d}{dq} \log \|Z\|_q \Big|_2 = -\frac{1}{4} S_e(ZZ^\dagger / z). \tag{8}$$

Indeed $\log \|Z\|_q = q^{-1} \log \sum_k \zeta_k^q$; substituting $\zeta_k^2 = z\mu_k$ in its derivative cancels the $\log z$ terms. Zero singular values contribute identically zero.

For $x, y > 0$, let $S_X = S_e(XX^\dagger/x)$, $S_Y = S_e(YY^\dagger/y)$. At $q = 2$, $r = \sqrt{x}$, $s = \sqrt{y}$, $r' = -rS_X/4$, and $s' = -sS_Y/4$. The derivative of the left side of (7) is $-S_e(MM^\dagger)$. The derivative of the right side is

$$(r+s)^2 \log(r+s) + (r-s)^2 \log|r-s| - xS_X - yS_Y = \ln 2 - W_e(|x-y|) - xS_X - yS_Y. \tag{9}$$

Since equality holds at 2, right differentiation preserves the inequality direction in (7). Rearranging gives

$$S_e(MM^\dagger) \geq W_e(|x-y|) + xS_X + yS_Y - \ln 2. \tag{10}$$

Block-diagonal entropy and the assumed weights give

$$xS_X + yS_Y = \ln 2 + \sum_{ij} p_{ij} S_e(M_{ij} M_{ij}^\dagger / p_{ij}). \tag{11}$$

Substitution and division by $\ln 2$ prove (5).

For clarity, without the balance assumption the same argument gives

$$S(MM^\dagger) \geq W(|x-y|) + \sum p_{ij} S(M_{ij} M_{ij}^\dagger / p_{ij}) + H(p_{00}, p_{01}, p_{10}, p_{11}) - h_2(x) - 1. \tag{12}$$

Balance is exactly what cancels the final three classical-entropy terms.

If $x = y$, differentiability of r, s gives $|r(q) - s(q)| = O(|q-2|)$, so the troublesome term in (7) has zero right derivative. If $x = 0$ or $y = 0$, M is block diagonal or block antidiagonal and (5) is the direct-sum entropy identity. Individual zero blocks contribute zero, with a continuous limit at fixed finite dimensions. These observations include all endpoints; no uniform limit over dimensions or tensor powers has been taken. □

3. Transverse spin correlations bound concurrence

Write the Bloch decomposition

$$\rho = \frac{1}{4} \left(I \otimes I + a \cdot \sigma \otimes I + I \otimes b \cdot \sigma + \sum_{ij} T_{ij} \sigma_i \otimes \sigma_j \right).$$

Choose any unit axes n, m for which a is parallel to n and b is parallel to m. If a Bloch vector is zero, its axis may be chosen arbitrarily. Let $U = n^\perp$, $V = m^\perp$, each a fixed two-dimensional real plane, and define

$$k_{n,m}(\rho) = \max_{u \in U, v \in V,\ |u|=|v|=1} |u^T T v|.$$

Lemma 2. For every two-qubit state and every such choice of axes,

$$\boxed{C(\rho) \leq k_{n,m}(\rho).} \tag{13}$$

The clause about fixed axes at zero Bloch vectors is part of the statement.

3.1 Rank at most two

A two-dimensional complex subspace of $\mathbb{C}^2 \otimes \mathbb{C}^2$ contains a product vector: represent two independent vectors as 2-by-2 matrices; the homogeneous quadratic determinant of their linear combination has a projective complex root. The corresponding nonzero matrix has rank one. A rank-at-most-two state therefore has a product vector in its kernel. Local unitaries put that vector at $|01\rangle$.

Positivity makes the 01 row and column vanish. Factor the state as $ZZ^\dagger$ with two columns (pad a rank-one factor with a zero column). A unitary on these columns makes the 00 row equal to $(L, 0)$, $L \geq 0$. If $L = 0$, the state is supported on $|1\rangle_A \otimes \mathbb{C}^2_B$, so its concurrence is zero and (13) is immediate. Otherwise the two local relative phases and the remaining second-column phase put the factor in the form

$$Z = \begin{pmatrix} L & 0 \\ 0 & 0 \\ u + iv & s \\ t & r \end{pmatrix}, \qquad L, s, t, r \geq 0, \quad u, v \in \mathbb{R}. \tag{14}$$

For example, the phases of s, t, r change by $\alpha + \gamma$, $\alpha + \beta$, $\alpha + \beta + \gamma$, respectively; these three independent combinations can set the nonzero entries nonnegative. This agrees with the three-qubit canonical purification form [4].

Keep the trace temporarily unnormalized:

$$N = L^2 + u^2 + v^2 + s^2 + t^2 + r^2, \qquad c = 2Lt.$$

The two potentially nonzero spin-flip roots are the singular values of

$$Z^T J Z = \begin{pmatrix} -c & -Lr \\ -Lr & 0 \end{pmatrix}. \tag{15}$$

This follows by the equality of the nonzero spectra of AB and BA, applied to $ZZ^\dagger J \overline{Z} Z^T J$. The singular-value difference of (15) is c. Thus $C(ZZ^\dagger/N) = c/N$.

Direct Pauli traces of $ZZ^\dagger$ give the unnormalized vectors

$$a = (2Lu, 2Lv, 2L^2 - N), \qquad b = (2(tu + sr), -2tv, N - 2(t^2 + r^2)), \tag{16}$$

and

$$T = \begin{pmatrix} c & 0 & a_x \\ 0 & -c & a_y \\ -b_x & -b_y & 2(L^2 + t^2 + r^2) - N \end{pmatrix} = D + a e_z^T - e_z b^T, \quad D = \mathrm{diag}(c, -c, N). \tag{17}$$

For $u_0 \in U$, $v_0 \in V$, both correction terms vanish, even when a Bloch vector is zero:

$$u_0^T T v_0 = u_0^T D v_0. \tag{18}$$

If $c = 0$, the claim is immediate. Otherwise D is invertible, and the two planes V and $D^{-1}U$ in $\mathbb{R}^3$ intersect nontrivially. Pick a unit vector v_0 in their intersection and set $u_0 = Dv_0/\|Dv_0\|$. Since

$$N \geq L^2 + t^2 \geq 2Lt = c,$$

the least singular value of D is c, so

$$u_0^T T v_0 = \|Dv_0\| \geq c. \tag{19}$$

After division by N, this proves the lemma for every rank-at-most-two state. The chosen axes can be carried through the local rotations, so no choice of a marginal eigenaxis has been lost.

3.2 Extension to all ranks

Work in the chosen local marginal eigenbases, and set $w = \rho_{00,11}$, $z = \rho_{01,10}$. Direct evaluation of the 2-by-2 transverse correlation matrix gives

$$k_{n,m}(\rho) = 2(|w| + |z|). \tag{20}$$

One way to see this is to use independent local diagonal phases to make $w, z \geq 0$; these phases act as orthogonal rotations within U, V. In that gauge the transverse matrix is $\mathrm{diag}(2(w+z), 2(z-w))$.

Consider the compact convex set $\mathcal{F}$ of density matrices whose local marginals are diagonal in these fixed bases, whose w, z are real, and for which $w, z \geq 0$. There are seven real affine equality constraints: trace one, four local off-diagonal marginal conditions, and two imaginary anti-diagonal conditions.

Suppose an extreme point in $\mathcal{F}$ has rank $q \geq 3$ and at least one of w, z positive. Hermitian perturbations supported on its range have $q^2 \geq 9$ real parameters. At most one nonnegativity constraint is active. Thus at most eight homogeneous linear conditions constrain such perturbations, leaving a nonzero perturbation H. For sufficiently small $\epsilon > 0$, both $\rho \pm \epsilon H$ remain positive on the support, preserve all equalities and active constraints, and preserve the inactive strict signs. They are distinct points of $\mathcal{F}$, contradicting extremality. Therefore every extreme point has rank at most two or satisfies $w = z = 0$.

It remains to check the zero face. Diagonal marginals imply its matrix is

$$\rho = \begin{pmatrix} \alpha & x & y & 0 \\ x^* & \beta & 0 & -y \\ y^* & 0 & \gamma & -x \\ 0 & -y^* & -x^* & \delta \end{pmatrix}. \tag{21}$$

Independent local diagonal phases make x, y real. For a real state, $R = \sqrt{\rho} J \sqrt{\rho}$ is real symmetric and the Wootters roots are the absolute values of its eigenvalues μ_j, because $\sqrt{\rho}\tilde{\rho}\sqrt{\rho} = R^2$. The triangle inequality gives

$$C(\rho) = \max(0, 2\max_j |\mu_j| - \sum_j |\mu_j|) \leq \left| \sum_j \mu_j \right| = |\operatorname{Tr}(\rho J)| = 0. \tag{22}$$

Thus the zero face has zero concurrence. The same separability also follows from the two-qubit PPT criterion.

Every member of a compact finite-dimensional convex set is a convex combination of its extreme points. On $\mathcal{F}$, concurrence is convex and $2(w+z)$ is linear. Applying the rank-at-most-two bound and the zero-face result to such a decomposition gives

$$C(\rho) \leq \sum_j p_j C(\rho_j) \leq \sum_j 2p_j(w_j + z_j) = 2(w+z) = k_{n,m}(\rho).$$

This proves Lemma 2 at every rank, including all degenerate marginal spectra. □

4. Closure to strong superadditivity and operational cost

Let $|\Psi\rangle$ be any pure state of $AA'BB'$, and let ρ_{AB} be its two-qubit marginal. Choose axes n, m as in Lemma 2. The unit spheres in the two planes are compact, so there are maximizing transverse vectors u, v. Reversing one sign if necessary gives

$$\kappa = u^T T v = k_{n,m}(\rho_{AB}) \geq C(\rho_{AB}) \geq 0.$$

Expand the qubits in the eigenbases of $u \cdot \sigma$ and $v \cdot \sigma$. Because $u \cdot a = v \cdot b = 0$, the local measurement distributions are both uniform. The four joint probabilities are exactly

$$p_{00} = p_{11} = \frac{1+\kappa}{4}, \qquad p_{01} = p_{10} = \frac{1-\kappa}{4}. \tag{23}$$

Across $AA' : BB'$, the coefficient matrix of $|\Psi\rangle$ has four A'-by-B' blocks with these squared norms. Let $|\phi_{ij}\rangle$ be the corresponding normalized nonzero block vectors. Lemma 1 and monotonicity of W give

$$S(\Psi_{AA'}) \geq W(\kappa) + \sum_{ij} p_{ij} S(\phi_{ij,A'}) \geq E_F(\rho_{AB}) + \sum_{ij} p_{ij} E(\phi_{ij}). \tag{24}$$

Tracing out AB shows

$$\rho_{A'B'} = \sum_{ij} p_{ij} |\phi_{ij}\rangle\langle\phi_{ij}|.$$

The definition of the convex roof therefore yields

$$S(\Psi_{AA'}) \geq E_F(\rho_{AB}) + E_F(\rho_{A'B'}). \tag{25}$$

Now take any pure decomposition $\Omega = \sum_\ell q_\ell |\Psi_\ell\rangle\langle\Psi_\ell|$. Apply (25) separately to each vector. Its measurement axes may depend on ℓ, because the resulting right-hand side consists of intrinsic marginal entanglement measures. Convexity gives

$$\sum_\ell q_\ell S(\Psi_{\ell,AA'}) \geq \sum_\ell q_\ell \{E_F(\rho^\ell_{AB}) + E_F(\rho^\ell_{A'B'})\} \geq E_F(\Omega_{AB}) + E_F(\Omega_{A'B'}).$$

Taking the infimum over decompositions proves (1). No optimal ensemble attainment assumption is necessary.

For $\Omega = \rho \otimes \sigma$, (1) is the lower bound in (2); tensor products of arbitrarily close-to-optimal ensembles give the upper bound. Iteration proves (3) for every finite n, including boundary states without a separate limiting argument. Finally the operational regularization theorem [3],

$$E_C(\rho) = \lim_{n\to\infty} \frac{E_F(\rho^{\otimes n})}{n},$$

proves (4). The source uses Bures error; the dimension-independent fidelity/trace-distance inequalities make vanishing Bures and trace error equivalent. □

5. Scope and physical meaning

The result concerns the entanglement needed asymptotically to prepare an ordinary two-qubit mixed state. Equation (3) rules out a reduction of the convex-roof cost through collective decompositions. Equation (4) identifies the Bell-pair formation rate. This concerns the asymptotic formation rate and does not imply reversibility with entanglement distillation.

The restriction on the peeled-off marginal in (1) is exactly 2×2. Arbitrary auxiliary dimensions are allowed, but the argument does not establish unrestricted-dimensional additivity. Splitting a large system into qubit pairs only bounds its entanglement by a sum of pair marginals, which can be strictly smaller than the entanglement of a grouped marginal. For crossed Bell pairs A_1B_2 and A_2B_1, the grouped formation entanglement is 2 bits while both same-index marginals have zero entanglement.

Fermionic superselection examples use a different convex roof and a different copy-composition convention. The ordinary-state interpretation of the example in [7] is separable (Eq. III.28).

References

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